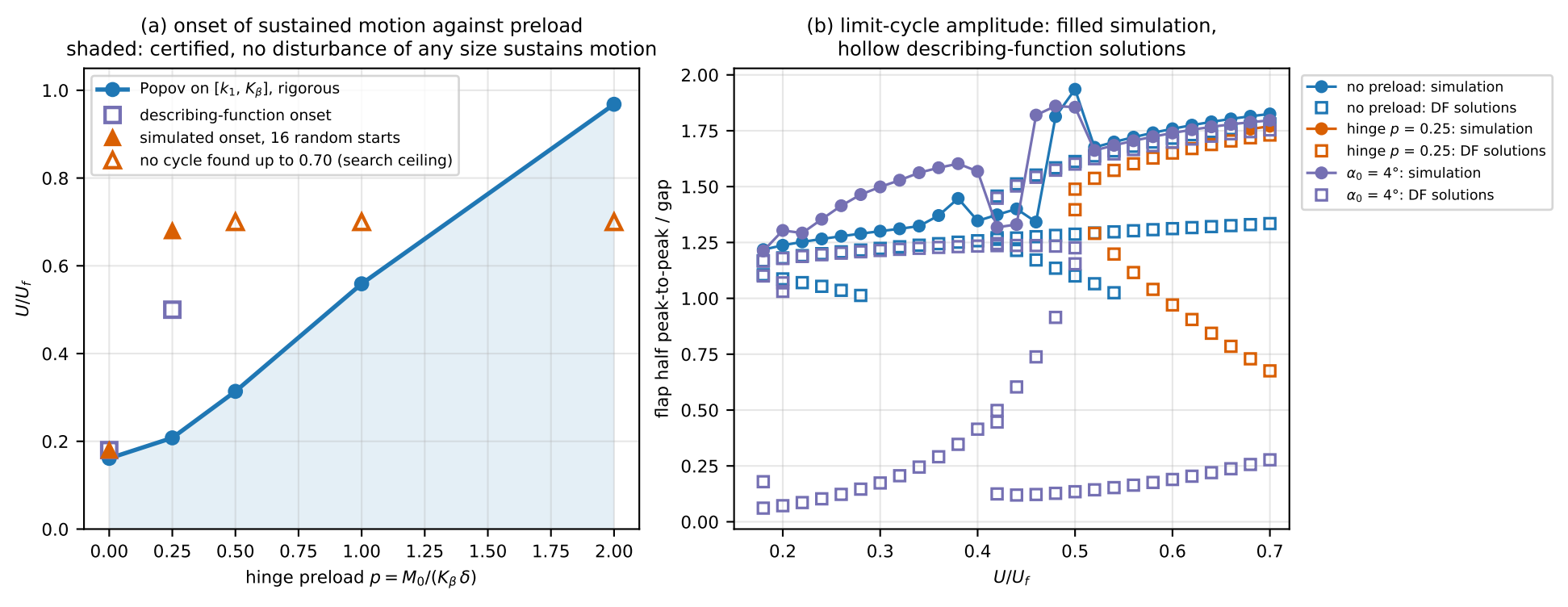
(a) onset of sustained motion against preload
shaded: certified, no disturbance of any size sustains motion
Popov on [k_1, K_β], rigorous
describing-function onset
simulated onset, 16 random starts
no cycle found up to 0.70 (search ceiling)
U/U_f
hinge preload $p = M_0/(K_\beta\,\delta)$
(b) limit-cycle amplitude: filled simulation,
hollow describing-function solutions
flap half peak-to-peak / gap
U/U_f
no preload: simulation
no preload: DF solutions
hinge p = 0.25: simulation
hinge p = 0.25: DF solutions
α_0 = 4°: simulation
α_0 = 4°: DF solutions